

Linear Algebra — Exercises

Linear mappings. Matrices of linear mappings

Exercise 1. Verify which of the following mappings are linear:

- (a) $F: \mathbb{R}^3 \rightarrow \mathbb{R}^2$, $F(x, y, z) = (x, z)$
- (b) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $F(x, y) = (3x + y + 1, 2x - 3y)$
- (c) $F: \mathbb{R}^2 \rightarrow \mathbb{R}^3$, $F(x, y) = (x + 2y, x - y, x)$
- (d) $F: \mathbb{R} \rightarrow \mathbb{R}$, $F(x) = |x|$
- (e) $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$, $F(x, y, z) = (xy, x, z)$

Exercise 2. Let $T: \mathbb{C} \rightarrow \mathbb{C}$, $T(z) = \bar{z}$.

- (a) Show that T is *not* linear if we consider $\mathbb{C}$ as a vector space over itself.
- (b) Show that T is linear if we consider $\mathbb{C}$ as a vector space over $\mathbb{R}$.

Exercise 3. Find $\ker F$, $\operatorname{im} F$, $\operatorname{r}(F)$ and the nullity of F for the following linear mappings $F: U \rightarrow V$:

- (a) $U = V = \mathbb{R}^3$, $F(x, y, z) = (x + y + z, x + y, x)$
- (b) $U = V = \mathbb{R}^3$, $F(x, y, z) = (x + y, x + y, x + 2y - z)$
- (c) $U = \mathbb{R}^3$, $V = \mathbb{R}^2$, $F(x, y, z) = (x - y + z, x + y - z)$

Exercise 4. Find the matrices of the linear mappings from Exercise 3 in the following bases of the spaces U and V , respectively:

- (a) the standard bases of unit vectors
- (b) $((1, 1, 1), (1, 1, 0), (1, 0, 0))$ and $((1, 0, 0), (1, 1, 0), (1, 1, 1))$
- (c) $((1, 1, 1), (1, 0, 0), (0, 0, 1))$ and $((1, 2), (3, 3))$

Exercise 5. Find a formula and the matrix in the standard basis for the linear operator F such that:

- (a) $F(1, 1, 1) = (1, 0, 1)$, $F(1, 1, 0) = F(0, 1, 1) = (1, 1, 1)$
- (b) $F(1, 2, 1) = (1, 0, 0)$, $F(1, 0, 1) = (0, 1, 0)$, $F(1, 0, 0) = (1, 0, 0)$
- (c) $F(1, 0, 0, 0) = F(0, 1, 0, 0) = F(0, 0, 1, 0) = F(0, 0, 0, 1) = (1, 1, 1, 1)$

Exercise 6. Assuming that $A = M_S(F)$ is the matrix of a linear operator F in the basis S , find the matrix $B = M_R(F)$ of F in the basis R :

- (a) $\begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$, for $S = ((1, 1), (1, 2))$ and $R = ((1, 0), (0, 1))$
- (b) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$, for $S = ((1, 1, 1), (0, 1, 2), (1, 0, 1))$ and $R = ((1, 0, 0), (0, 1, 0), (0, 0, 1))$
- (c) $\begin{pmatrix} 1 & 2 & 2 \\ 1 & 1 & 1 \\ 2 & 0 & 0 \end{pmatrix}$, for $S = ((1, 1, 1), (0, 1, 2), (1, 0, 1))$ and $R = ((1, 0, 0), (0, 1, 0), (0, 0, 1))$
- (d) $\begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$, for $S = ((1, 0, 1, 0), (0, 0, 1, 1), (1, 1, 0, 0), (1, 0, 1, 1))$ and $R = ((1, 1, 1, 0), (1, 1, 0, 1), (1, 0, 1, 1), (0, 1, 1, 1))$

Exercise 7. For each pair of matrices A and B from Exercise 6, find a matrix P such that $A = P^{-1} \cdot B \cdot P$. Verify your solution by matrix multiplication.